# RTH mistakes - proof report

PClaimAssist research tests, test 8, supplementary to Eq. 3.12 (documentation error reduction) - results of 2026-11-06 06:33:39, report generated 2026-11-06

**Result.** The clinic's common return-to-hospital mistakes were planted **278** times on the real forms page. PClaimAssist **prevented 184**, **blocked 44** (Save as PDF refuses), **warned on 26**, and **let 24 through**. Error reduction against manual typing: **82.0%** counting a warning as an error left, **91.4%** counting a warning as caught. Weighted by how common the clinic says each reason is: **72.5%** and **93.1%**.

**Read with care.** Manual = each form typed separately with nothing checking it, so every planted mistake stays in the claim: that is the assumption of the comparison, not a measurement of staff. The result shows whether the system stops a mistake once it is made, not how often staff make mistakes; Eq. 3.12 proper comes from the user sessions. The weights are the clinic's estimate, not counted from returned claims.

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The PDF also has bookmarks, and every page links back to these contents.

# 1. Result at a glance

| Outcome | Planted mistakes | Share | What it means |
|---|---|---|---|
| Prevented | 184 | 66.2% | The second copy cannot differ: locked box filled from Patient details, or one value for every form |
| Blocked | 44 | 15.8% | A critical rule fires: Save as PDF refuses |
| Warned | 26 | 9.4% | A warning fires: staff are told, the PDF can still be saved |
| Gets through | 24 | 8.6% | In the claim, and nothing notices |
| All planted mistakes | 278 | 100.0% | |

| Error reduction | Warning = error left (strict) | Warning = caught |
|---|---|---|
| All planted mistakes (each counts once) | 82.0% | 91.4% |
| Weighted by how common each reason is | 72.5% | 93.1% |

Error reduction % = (manual errors - PClaimAssist errors) / manual errors x 100, the form of Eq. 3.12. Weights: very common 4, common 3, slightly uncommon 2, uncommon 1.

## 2. Per RTH reason

| RTH reason | How common | Planted | Prevented | Blocked | Warned | Gets through | Reduction (strict) | Reduction (warning = caught) |
|---|---|---|---|---|---|---|---|---|
| Dates | very common | 82 | 54 | 22 | 0 | 6 | 92.7% | 92.7% |
| Gravida and baby details | slightly uncommon | 14 | 0 | 0 | 14 | 0 | 0.0% | 100.0% |
| PIN | slightly uncommon | 54 | 36 | 18 | 0 | 0 | 100.0% | 100.0% |
| Inconsistent admission reason | uncommon | 12 | 8 | 4 | 0 | 0 | 100.0% | 100.0% |
| Name | very common | 88 | 70 | 0 | 0 | 18 | 79.5% | 79.5% |
| Clinical details and examinations | common | 28 | 16 | 0 | 12 | 0 | 57.1% | 100.0% |
| All planted mistakes | | 278 | 184 | 44 | 26 | 24 | 82.0% | 91.4% |

## 3. Per mistake

Each mistake was planted in every test patient it applies to. *Rules* are the rules that fired; *How* is what the page did.

| ID | Mistake | How it happens | Planted | Outcome | Rules | How |
|---|---|---|---|---|---|---|
| D1 | Date admitted typed one day off on one form | Dual entry | 18 | **Prevented 18** | none | CF2's box is locked: it is filled from Patient details, the same value on every form |
| D2 | Date discharged typed differently on one form | Dual entry | 18 | **Prevented 18** | none | CF2's box is locked: it is filled from Patient details, the same value on every form; CF4's box is locked: it is filled from Patient details, the same value on |
| D3 | Patient's birth date with a wrong year on one form | Dual entry | 18 | **Prevented 18** | none | CF1's box is locked: it is filled from Patient details, the same value on every form |
| D4 | Day and month swapped in the patient's birth date | Typed once | 6 | **Gets through 6** | none | Gets through |
| D5 | Discharge date before the admission date | Typed once | 18 | **Blocked 18** | IC-03, IC-05, IC-08, IC-11 | Blocked |
| D6 | Delivery date outside the stay | Typed once | 4 | **Blocked 4** | IC-05, IC-08 | Blocked |
| G1 | Gravida in CF4's OB history differs from the GRAVIDA in the CF2 admission diagnosis | Dual entry | 4 | **Warned 4** | IC-10 | Warned |
| G2 | Para in CF4's OB history differs from the PARA in the admission diagnosis | Dual entry | 4 | **Warned 4** | IC-10 | Warned |
| G3 | Baby's claim gives the mother a different gravida/para than the mother's claim | Dual entry | 2 | **Warned 2** | IC-13 | Warned |
| G4 | Baby's birth date differs from the mother's delivery date | Dual entry | 2 | **Warned 2** | IC-13 | Warned |
| G5 | Baby's sex or birth weight differs between the mother's and the baby's claim | Dual entry | 2 | **Warned 2** | IC-13 | Warned |
| P1 | PIN with one wrong digit on one form | Dual entry | 18 | **Prevented 18** | none | CF1's box is locked: it is filled from Patient details, the same value on every form |
| P2 | PIN with two digits swapped on one form | Dual entry | 18 | **Prevented 18** | none | CF1's box is locked: it is filled from Patient details, the same value on every form; CF4's box is locked: it is filled from Patient details, the same value on |
| P3 | PIN with a digit missing (11 digits) | Typed once | 18 | **Blocked 18** | IF-01 | Blocked |
| A1 | CF2 admission diagnosis differs from CF4 admitting diagnosis | Dual entry | 8 | **Prevented 8** | none | CF4's box is locked: it is filled from Patient details, the same value on every form |
| A3 | CF2 final diagnosis differs from CF4 discharge diagnosis | Dual entry | 4 | **Blocked 4** | IC-09 | Blocked |
| N1 | Name spelled with one letter different on one form | Dual entry | 18 | **Prevented 18** | none | CF2's box is locked: it is filled from Patient details, the same value on every form; CF4's box is locked: it is filled from Patient details, the same value on |
| N2 | Middle name left out on one form | Dual entry | 16 | **Prevented 16** | none | CF2's box is locked: it is filled from Patient details, the same value on every form |
| N3 | First and last name swapped on one form | Dual entry | 18 | **Prevented 18** | none | CF2's box is locked: it is filled from Patient details, the same value on every form |
| N4 | Suffix left out on one form | Dual entry | 18 | **Prevented 18** | none | CF1's box is locked: it is filled from Patient details, the same value on every form |
| N5 | Name misspelled the same way on every form | Typed once | 18 | **Gets through 18** | none | Gets through |
| C2 | ICD-10 or RVS case rate code differs between CSF and CF4 | Dual entry | 8 | **Prevented 8** | none | typing on CF4 changed every form (one value) |

| | | | | | | |
|---|---|---|---|---|---|---|
| C3 | Attending midwife's name or accreditation number differs on one form | Dual entry | 8 | **Prevented 8** | none | typing on CF4 changed every form (one value) |
| C4 | Age of gestation in the diagnosis does not fit the LMP on CF4 | Typed once | 4 | **Warned 4** | IC-12 | Warned |
| C6 | Course in the ward dated outside the stay | Typed once | 8 | **Warned 8** | IC-11 | Warned |

## 4. What still gets through

These mistakes are in the claim and no check notices them. They are limits of the system.

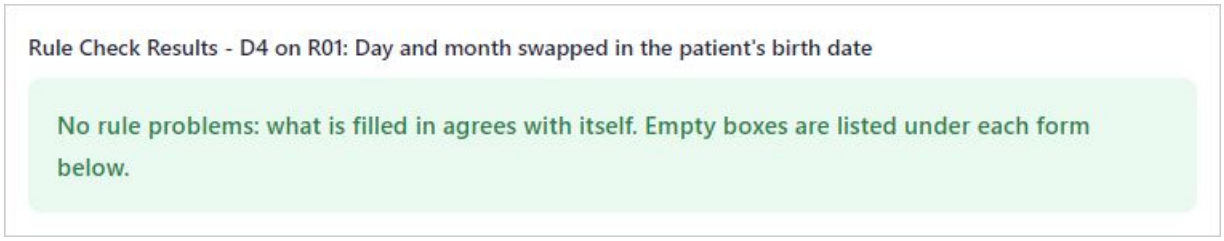
| ID | Mistake | Planted | Gets through | Why |
|---|---|---|---|---|
| D4 | Day and month swapped in the patient's birth date | 6 | 6 | A swapped day and month is still a real date; nothing else on the forms says which is right. |
| N5 | Name misspelled the same way on every form | 18 | 18 | The forms agree with each other; only the patient's ID shows the right spelling. |

## 5. Proof: the page's own rule panel

For each mistake that was not prevented, the page's own "Rule Check Results" panel after the mistake was made, on the first patient it was planted in (every picture is in outputs/rth_errors/proof/). Prevented mistakes have no panel: the box could not be typed differently (see section 3).

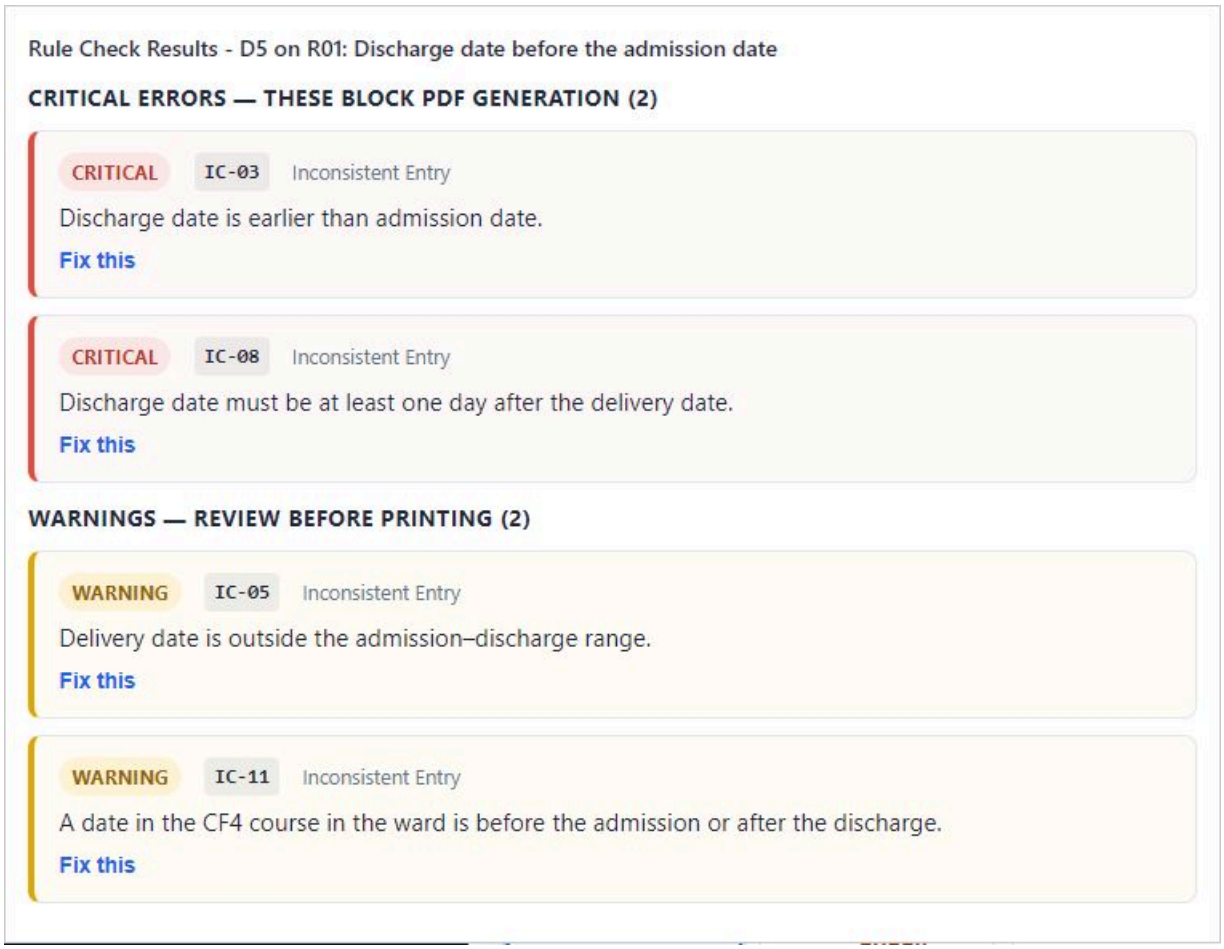
### D4 on R01 - Day and month swapped in the patient's birth date: Gets through

typed '1993-09-01' in Patient details patientDOB



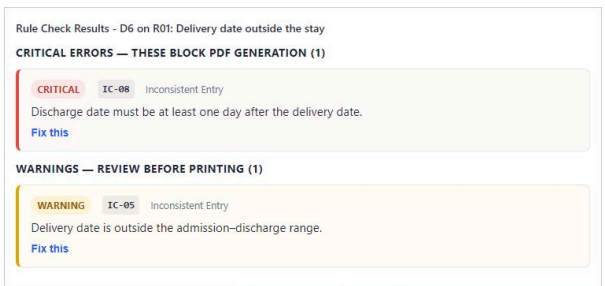

### D5 on R01 - Discharge date before the admission date: Blocked (IC-03, IC-05, IC-11, IC-08)

typed '2026-08-27' in Patient details dateDischarge



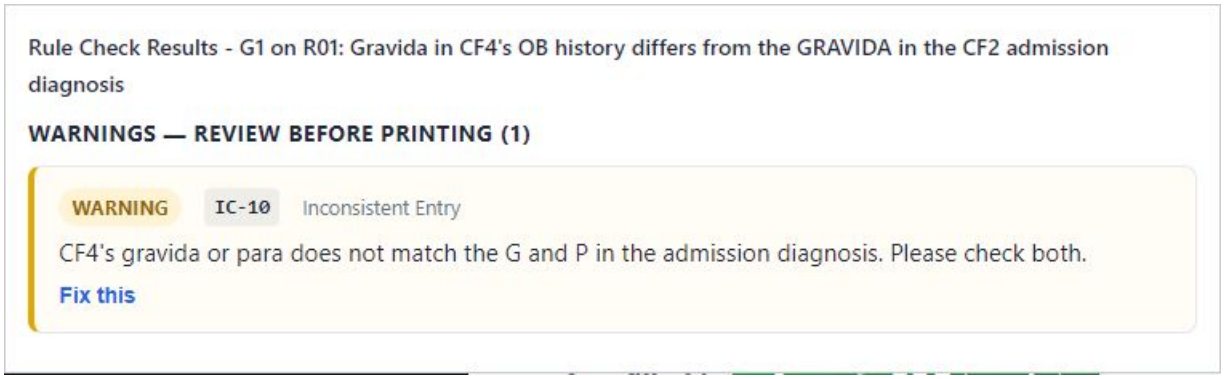

### D6 on R01 - Delivery date outside the stay: Blocked (IC-05, IC-08)

typed '2026-09-03' in Patient details deliveryDate



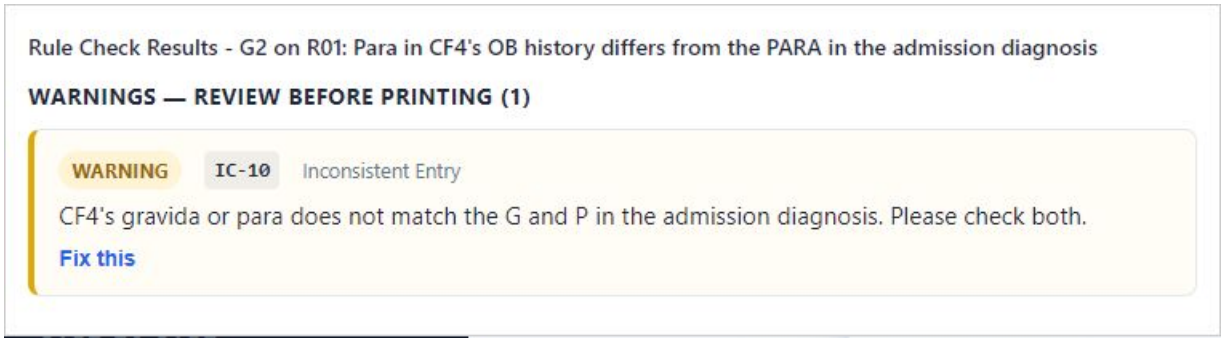
## G1 on R01 - Gravida in CF4's OB history differs from the GRAVIDA in the CF2 admission diagnosis: Warned (IC-10)

typed '4' in CF4 gravida (typed)



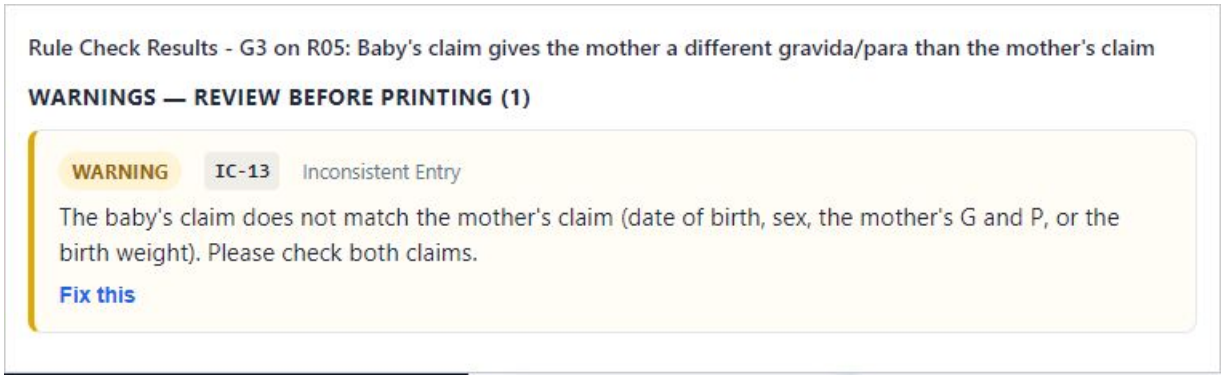

## G2 on R01 - Para in CF4's OB history differs from the PARA in the admission diagnosis: Warned (IC-10)

typed '4' in CF4 para (typed)



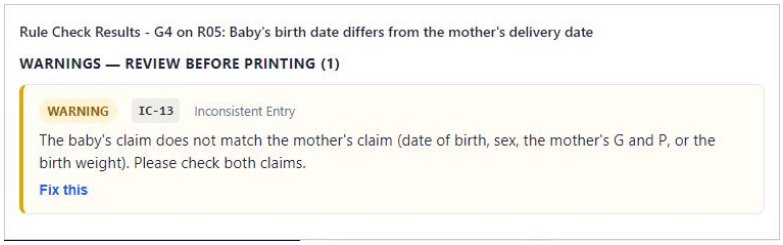

## G3 on R05 - Baby's claim gives the mother a different gravida/para than the mother's claim: Warned (IC-13)

typed 'LIVE TERM BABY GIRL AGA DELIVERED VIA NSVD TO A G4P4 MOTHER AS OF 8,9' in the baby's claim, Patient details admissionDx | typed 'LIVE TERM BABY GIRL AGA DELIVERED VIA NSVD TO A G4P4 MOTHER WEIGHING 2800 GRAMS W/ AS OF 8,9' in the baby's claim, Patient details dischargeDx | typed 'LIVE TERM BA



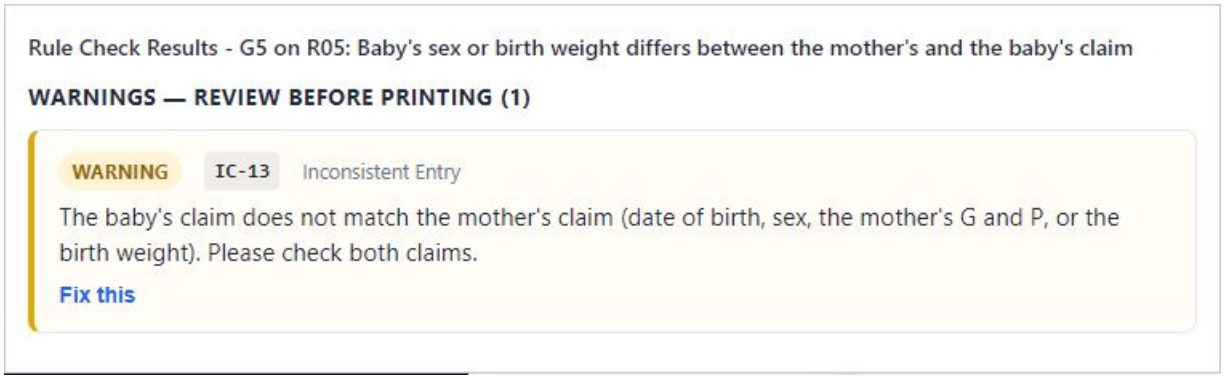

## G4 on R05 - Baby's birth date differs from the mother's delivery date: Warned (IC-13)

typed '2026-08-30' in the baby's claim, Patient details patientDOB



## G5 on R05 - Baby's sex or birth weight differs between the mother's and the baby's claim: Warned (IC-13)

typed 'Male' in the baby's claim, Patient details patientSex



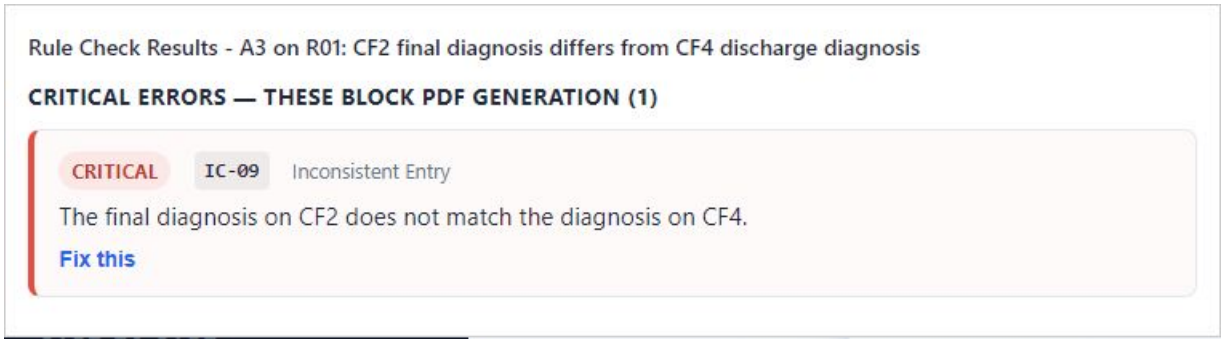

## P3 on R01 - PIN with a digit missing (11 digits): Blocked (IF-01)

typed '99-12346461-9' in Patient details memberPIN | typed '99-12346461-9' in Patient details patientPIN



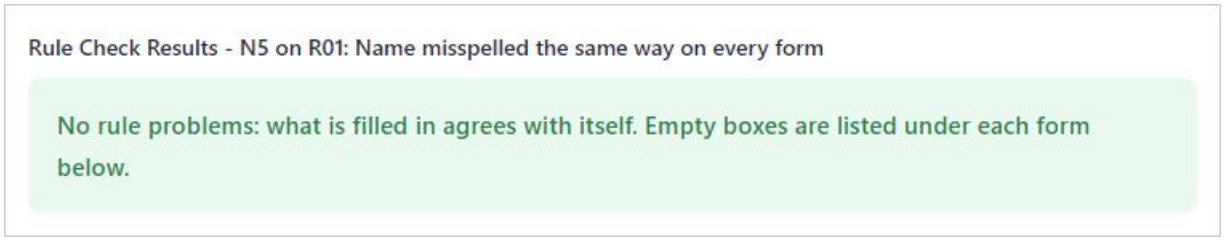

## A3 on R01 - CF2 final diagnosis differs from CF4 discharge diagnosis: Blocked (IC-09)

typed 'PUFT 37 4/7 WKS AOG DELIVERED VIA LOW SEGMENT CESAREAN SECTION TO A LIVE TERM BABY GIRL AGA W/ AS OF 8,9 G3P3 (3003)' in CF4 cf4FinalDiagnosis (typed)




## N5 on R01 - Name misspelled the same way on every form: Gets through

typed 'BANDOLIB' in Patient details patientLastName



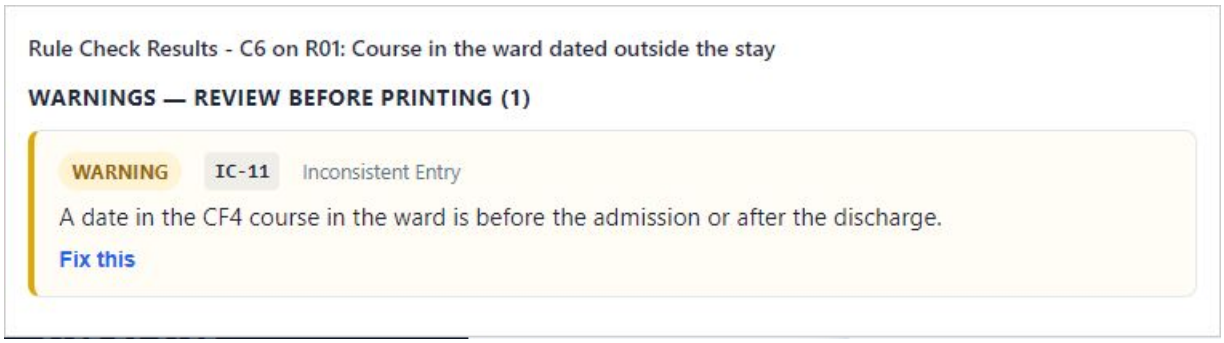

## C4 on R01 - Age of gestation in the diagnosis does not fit the LMP on CF4: Warned (IC-12)

typed '2026-01-10' in Patient details lmp



## C6 on R01 - Course in the ward dated outside the stay: Warned (IC-11)

typed '2026-09-02' in CF4 cf4CourseDate1 | typed 'VITAL SIGNS TAKEN' in CF4 cf4CourseOrder1



## 6. Listed but not counted

| ID | RTH reason | Mistake | Why not counted | Detail |
|---|---|---|---|---|
| C1 | Clinical details and examinations | History of present illness on CF4 differs from the brief history | Printed on one form only | The history of present illness is printed on CF4 only; Patient details' brief history is not printed. There is no second copy on the forms to differ. Printed on: cf4HistoryPresentIllness: CF4; briefHistory: no form |
| C5 | Clinical details and examinations | Physical examination does not fit the diagnosis | Printed on one form only | The physical examination is printed on CF4 only (one form): there is no second copy on the forms to differ. Printed on: cf4Abdomen: CF4; cf4AbdomenOthers: CF4; cf4GeneralSurvey: CF4 |
| C7 | Clinical details and examinations | Drugs given do not fit the delivery | Printed on one form only | The drugs given are printed on CF4 only (one form): there is no second copy on the forms to differ. Printed on: cf4DrugName1: CF4; cf4DrugName2: CF4; cf4DrugName3: CF4 |
| C8 | Clinical details and examinations | Vital signs in CF4's boxes differ from the vital signs written in the physical examination or the course in the ward | Printed on one form only | The vital signs are printed in CF4's boxes only; the physical examination text and the course in the ward do not repeat them on the forms. A mismatch with the clinic's chart is outside the forms and is not checked. Printed on: cf4VitalBP: CF4; cf4VitalHR: CF4; cf4VitalRR: CF4; cf4VitalTemp: CF4 |
| A2 | Inconsistent admission reason | CF4 chief complaint does not fit the admission diagnosis | Left out (Include = no) | Not a mistake: the chief complaint is always different from the admission diagnosis (the clinic) |

## 7. What this means

**Typing a value once stops the dual-entry mistakes.** Names, PIN, dates and the admission diagnosis are typed once on Patient details and locked on every form, so a second copy cannot differ; the case rate codes and the attending midwife copy to every form. Impossible values (an 11-digit PIN, a discharge before the admission, a delivery outside the stay, a cesarean on CF4 against a normal delivery on CF2) block Save as PDF. The OB score, the course in the ward, the LMP and a baby's claim against its mother's are warned on.

Of 278 planted mistakes, 228 were stopped and 26 warned on; 24 got through: a value that is wrong the same way everywhere (a misspelled name, a swapped day and month). Those need the patient's ID or record to catch, which the forms do not have.